

COURSE INFORMATION

DSK808 Data Visualization

Fall 2025

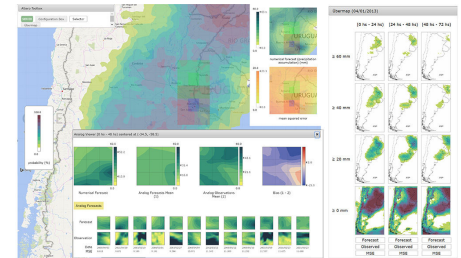
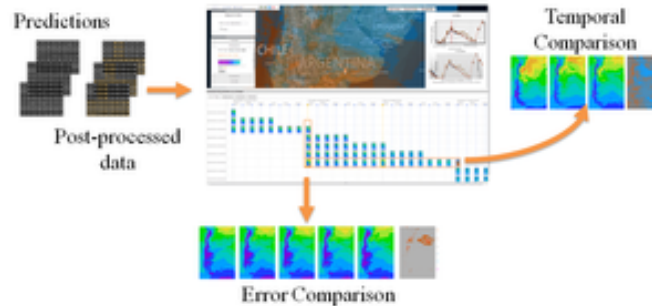


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Alexandra Diehl

- Associate Professor of Data Science at IMADA, SDU
- I work in the areas of data science, visualization, visual analytics, high-dimensional visualization, geovisualization, and best practices and guidelines for visualization.
- Intro to Programming and Data Visualization



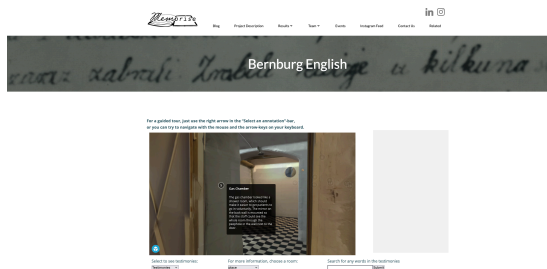
Get to know your TA 😊



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Camilla Vang Østergaard

- Ph.D. Student at Center for Visual Data Science at IMADA, SDU
- My work focuses on data science, visualization, and 3D models, primarily in the field of digital humanities.
- I work on the Memorise project, a European project whose goal is to communicate the heritage of Nazi persecution in new digital ways.



<https://memorise.sdu.dk/bernburg-3d-model/>

ORGANIZATION

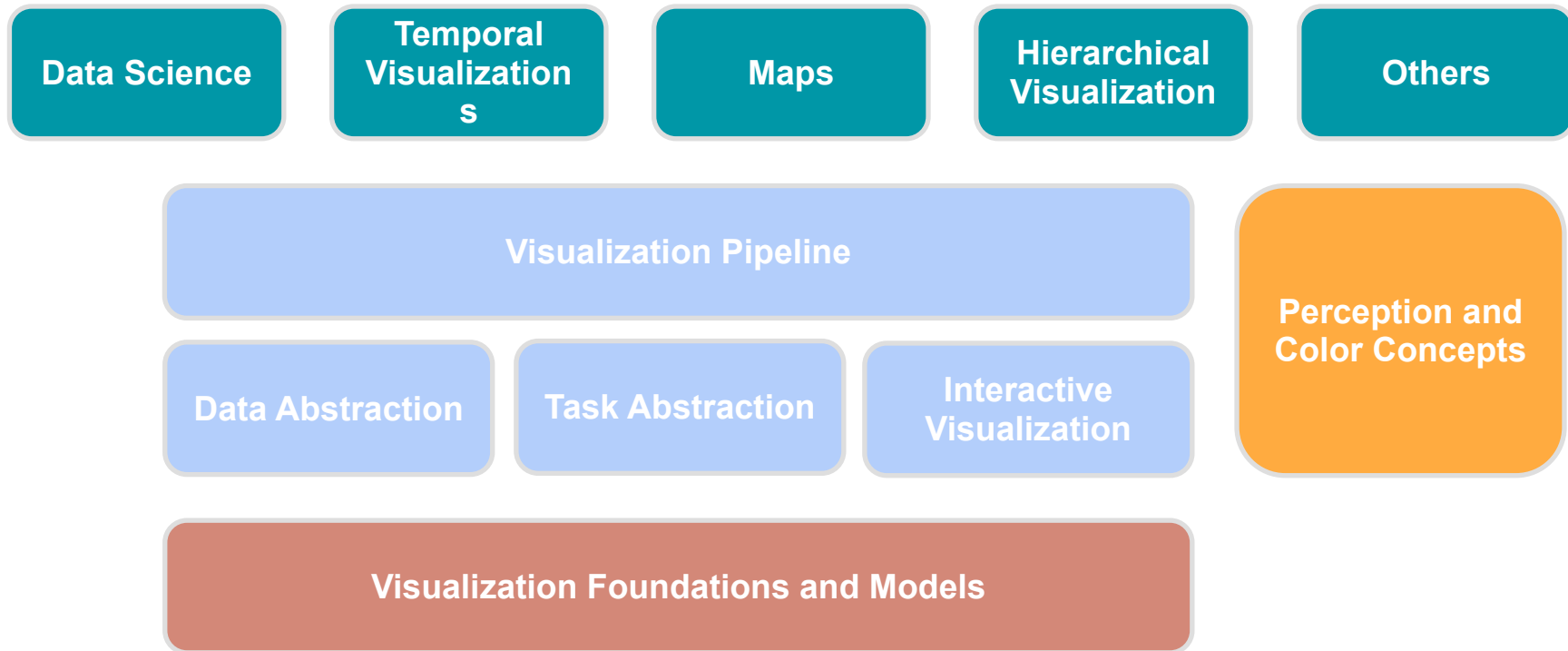
DSK808 Data Visualization

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Learning Objectives

- Apply and explain visual design methods for arbitrary user tasks
- Choose appropriate visual characteristics for mapping data features
- Apply and explain appropriate advanced visualization methods
- Evaluate the quality of and suggest improvements to visual mappings
- Team working: coming with strategies and solving visual design tasks together as a team.

CONTENT



CONTENT

- **Visualization Foundations and Models:** Explore core principles and frameworks that guide effective visual communication.
- **Data Abstraction:** Practice cleaning, transforming, and structuring data for visual analysis.
- **The Visual Pipeline:** Map data attributes to visual channels like position, color, and size using encoding strategies.
- **Tasks Abstraction:** Design visualizations tailored to user goals such as comparison, filtering, and exploration.
- **Perception and Color:** Apply perceptual principles and color theory to enhance clarity and accessibility.
- **Interactive Visualization:** Build dynamic interfaces with zoom, filter, hover, and animation to deepen user engagement.
- **Data Science Visualizations:** daily life data scientists visualizations.
- **Temporal Visualization:** Create time-series plots, timelines, and animated transitions for temporal data.
- **Maps:** Use geospatial tools to build choropleths, point maps, and spatial overlays.
- **Hierarchical Visualization:** Represent nested data using tree maps, sunbursts, and dendrograms.
- **Other Visualizations:** Explore other visualizations for complex data scenarios.

Syllabus

WEEK	SESSION	DATE	TITLE
36	1	Tue 02/09/2025	Visualization Foundations and Models
37	2	Tue 09/09/2025	Data Abstraction /Basics of Data Processing (brush-up)
38	3	Tue 16/09/2025	Visual Pipeline and Visual Encodings
39	4	Tue 23/09/2025	Tasks Abstraction
40	5	Tue 30/09/2025	Perception and Color
41	6	Tue 07/10/2025	Interactive Visualization
42		Tue 14/10/2025	No theory lecture
43	7	Tue 21/10/2025	Visualization Techniques: Data Science
44	8	Tue 28/10/2025	Visualization Techniques: Temporal Vis
45		Tue 04/11/2025	No theory lecture

Syllabus

WEEK	SESSION	DATE	TITLE
46	16	Tue 11/11/2025	Visualization Techniques: Maps
47	18	Tue 18/11/2025	Visualization Techniques: Hierarchical Vis
48	20	Tue 25/11/2025	Visualization Techniques: Others

Lab Exercises

What to expect

- Code examples with Dash/Plotly
- Presentation of the exam and formation of exam groups
- Examples of data visualizations
- Examples of data handling/cleaning

The first three sessions will be class teaching. The remaining sessions are dedicated to project-specific support, where groups can book individual time slots for guidance.

Lectures

Help with projects

Tue 02/09/2025

Tue 09/09/2025

Tue 16/09/2025

Tue 23/09/2025

Tue 30/09/2025

Tue 07/10/2025

Tue 14/10/2025

Tue 21/10/2025

Tue 28/10/2025

Tue 04/11/2025

Tue 11/11/2025

Tue 18/11/2025

Tue 25/11/2025

Tue 02/12/2025

Tue 09/12/2025

Tue 16/12/2025

Projects & Portfolio Exam

Group project: up to 4 students

▼ Additional information

The portfolio exam consists of the following elements:

- a group project with a written scientific short paper (2-4 pages) describing the group's visualization project
- a (short) demonstration video of the visualization as supplemental material to the paper
- a shared presentation of the visualization project with an oral group discussion
- short individual sessions with questions on theoretical contents of the course

Portfolio exam in January

- Prerequisite: successful group project
- poster sessions + short individual sessions (individual grade)

Generative AI (ChatGPT, ..etc)

What are the general rules?

It is permitted for students to use generative AI when completing written take-home assignments, including bachelor projects, graduation projects and master's theses, provided that generative AI content is clearly indicated.

It is not permitted to use generative AI in connection with written site tests, other supervised exams or in connection with oral tests.

Any exceptions will be stated in the course description.

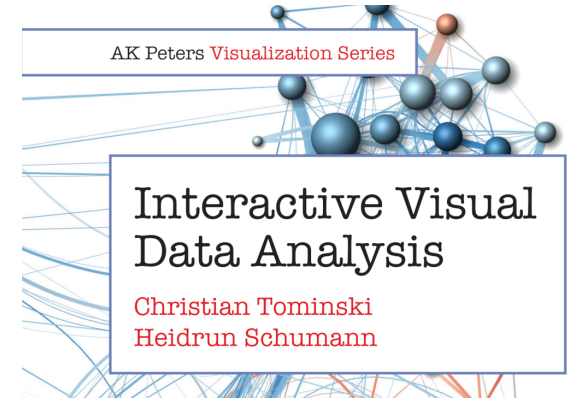
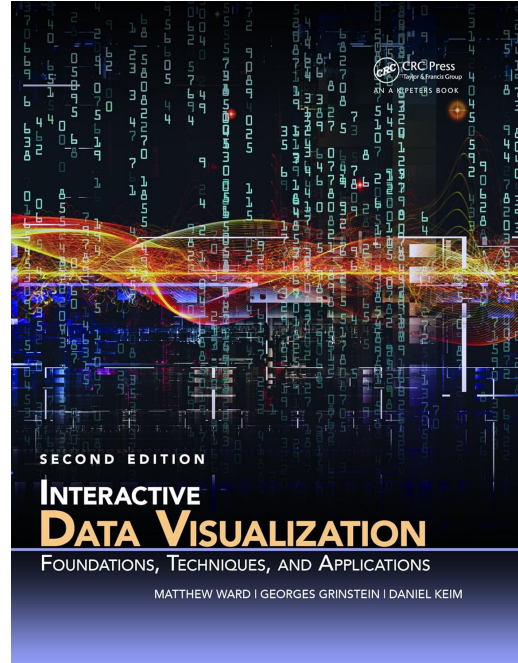
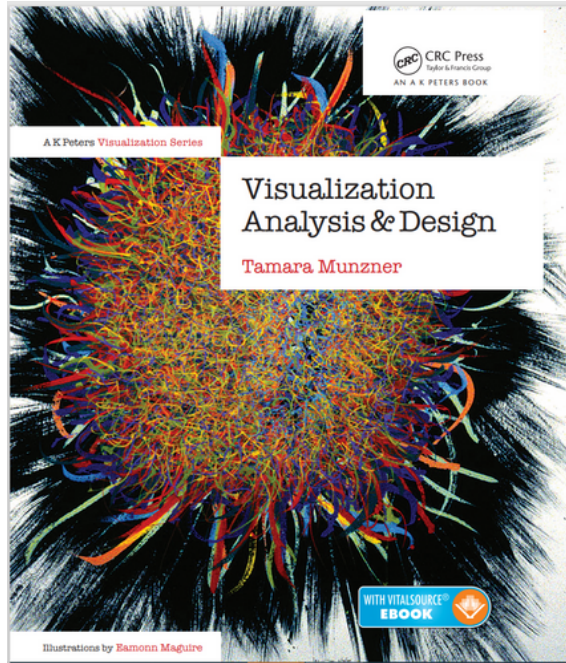


https://sdunet.dk/en/undervisning-og-eksamen/nyheder/2024/0628_gai

7-point grading scale

- 3** is given for a performance **unacceptable** in all aspects and means that you have **failed** the exam.
- 00** reflects a performance that does not meet the minimum requirements for passing.
- 02** means a performance just meeting the requirements for passing.
- 4** reflects a **fair performance** that still has several weaknesses.
- 7** is equal to a **good performance**, though still with some weaknesses
- 10** is consistent with a **very good performance**, which only has minor weaknesses.
- reflects an containing only few or no weaknesses at all.

References



QUESTIONS?

